

03.-Conectando-tres-contenedores

En esta actividad se conectan tres contenedores dentro de una misma red personalizada, integrándolos entre si mediante sus direcciones IP internas.

Se utilizaron las siguientes imágenes:

- eclipse-mosquitto:2.0.18
- grafana/grafana:11.3.0
- apache/kafka:3.8.

Creación de la red

docker network create examen_curs_00568

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
10f57f843433	bridge	bridge	local
5b2b9632ea62	examen_curs_00568	bridge	local
7cfb94847053	host	host	local
8d0e766d90b6	none	null	local

Creación de los contenedores, uniéndolos a la red examen_curs_00568

```
docker container run -d \  
-p 3000:3000 \  
--name DOCKER-CONTAINER-V1 \  
--network examen_curs_00568 \  
grafana/grafana:11.3.0
```

```
docker container run -d \  
-p 9092:9092 \  
--name DOCKER-CONTAINER-KAFKA \  
--network examen_curs_00568 \  
apache/kafka:3.8.0
```

```
docker container run -d \  
-p 1883:1883 \  
--name DOCKER-CONTAINER-MOSQUITTO \  
--network examen_curs_00568 \  
eclipse-mosquitto:2.0.18
```

Inspeccionamos la red examen_curs_00568

docker network inspect examen_curs_00568

```
Containers": {
  "a7e8111a57f248a9c255a3c41a4d44e8753127115aeddb143bc9369e05b80564": {
    "Name": "DOCKER-CONTAINER-V1",
    "EndpointID": "29d14ccce7fbd0fe8a2bd7342a8831bf3350b3fac99baf939731523e70642632",
    "MacAddress": "ce:4f:ed:6f:85:fc",
    "IPv4Address": "172.18.0.2/16",
    "IPv6Address": ""
  },
  "c3c99261671a8e17bdec445e1a81128a64751d70f9ab63bd355fac0da904ee24": {
    "Name": "DOCKER-CONTAINER-KAFKA",
    "EndpointID": "7184b9ed17c37de0d2c96fb71b534d3f6b92bf3ce2192f3b7e99a71faf642a3d",
    "MacAddress": "1e:f2:23:d9:9d:ca",
    "IPv4Address": "172.18.0.3/16",
    "IPv6Address": ""
  },
  "c59ab0763f7a90d50258f3ed94a1f4370fe7caa3e9831c6d228f49f8914e5ce2": {
    "Name": "DOCKER-CONTAINER-MOSQUITO",
    "EndpointID": "4e334f5d3de695f204c888d9777c077ef07fca5ba81520a7cc122ccf42b78393",
    "MacAddress": "0e:dd:bb:8f:db:96",
    "IPv4Address": "172.18.0.4/16",
    "IPv6Address": ""
  }
}
```

Para simplificar la lectura de la salida, se creó una función personalizada (docker-net-containers, documentada en el Docker Sheet del componente 1) que **filtra únicamente el nombre y la IP de cada contenedor conectado a la red**:

docker-net-containers examen_curs_00568

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker-net-containers examen_curs_00568
DOCKER-CONTAINER-V1 -> 172.18.0.2/16
DOCKER-CONTAINER-KAFKA -> 172.18.0.3/16
DOCKER-CONTAINER-MOSQUITO -> 172.18.0.4/16
```

Pruebas de comunicación entre contenedores

Realizamos las pruebas de comunicación entre contenedores de la misma red

docker exec -it DOCKER-CONTAINER-V1 ping -c 4 172.18.0.3

docker exec -it DOCKER-CONTAINER-KAFKA ping -c 4 172.18.0.4

```
DOCKER-CONTAINER-V1 -> 172.18.0.2/16
DOCKER-CONTAINER-KAFKA -> 172.18.0.3/16
DOCKER-CONTAINER-MOSQUITO -> 172.18.0.4/16

jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker exec -it DOCKER-CONTAINER-V1 ping -c 4 172.18.0.3
docker exec -it DOCKER-CONTAINER-V1 ping -c 4 172.18.0.4
PING 172.18.0.3 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=42 time=0.304 ms
64 bytes from 172.18.0.3: seq=1 ttl=42 time=0.093 ms
64 bytes from 172.18.0.3: seq=2 ttl=42 time=0.066 ms
64 bytes from 172.18.0.3: seq=3 ttl=42 time=0.114 ms

--- 172.18.0.3 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.066/0.144/0.304 ms
PING 172.18.0.4 (172.18.0.4): 56 data bytes
64 bytes from 172.18.0.4: seq=0 ttl=42 time=0.229 ms
64 bytes from 172.18.0.4: seq=1 ttl=42 time=0.073 ms
64 bytes from 172.18.0.4: seq=2 ttl=42 time=0.070 ms
64 bytes from 172.18.0.4: seq=3 ttl=42 time=0.070 ms

--- 172.18.0.4 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.070/0.110/0.229 ms
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker exec -it DOCKER-CONTAINER-KAFKA ping -c 4 172.18.0.4
PING 172.18.0.4 (172.18.0.4): 56 data bytes
64 bytes from 172.18.0.4: seq=0 ttl=42 time=0.207 ms
64 bytes from 172.18.0.4: seq=1 ttl=42 time=0.075 ms
64 bytes from 172.18.0.4: seq=2 ttl=42 time=0.074 ms
64 bytes from 172.18.0.4: seq=3 ttl=42 time=0.067 ms

--- 172.18.0.4 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.067/0.105/0.207 ms
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$
```